

CSE437 Final Project Report

Disparity and Error Analysis in Linear Income Classification

An Audit of 2023 Texas ACS PUMS Microdata

Course & Semester: CSE437 Data Science, Summer 2026

Group Members

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GitHub Repository: <https://github.com/tanjilaafsarirubina/cse437-income-disparity-14>

Dataset (Google Drive):

<https://drive.google.com/drive/folders/1E6GYPV0siUHCq2ohG6EdfZky0AJJOXd3?usp=sharing>

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Summary

This study evaluates demographic and employment-related predictive disparities in linear income classification using the 2023 U.S. Census Bureau American Community Survey (ACS) 1-Year Public Use Microdata Sample (PUMS) for Texas. The objective is to classify whether a full-time employed civilian worker belongs to the top earnings quartile (\$90,000 threshold), creating the binary target variable HIGH_EARNER. We benchmark two linear classification model families: a Support Vector Classifier (LinearSVC with L2 regularization) and LogisticRegression across 28 engineered demographic, educational, and occupational features. On the held-out test partition of 6,000 records, the tuned LinearSVC achieved our headline metric of an F1-score of 57.71% (80.58% overall accuracy, 67.72% precision, 50.28% recall), substantially outperforming the majority-class baseline (0.00% F1). Our primary finding reveals systematic subgroup erasure: the linear decision boundary predicts a 0.00% high-earner rate for women lacking post-secondary degrees, despite an empirical 8.29% baseline in the test cohort ($z = -10.70$, $p < 0.001$, exact binomial $p = 2.50 \times 10^{-48}$). This confirms that standard linear decision boundaries severely amplify empirical wage disparities for lower-credentialed demographic subsets.

1. Problem and Dataset

1.1 Problem Statement

Algorithmic classification models deployed in labor analytics, credit scoring, and public resource allocation frequently use linear decision boundaries under the assumption of mathematical fairness and interpretability. However, when linear hyperplanes partition complex socioeconomic microdata, additive feature weights can interact with historical wage distributions to create severe localized disparities. This study models top-quartile income classification to investigate where and why linear models fail across demographic intersections (gender, educational attainment, and age) and institutional employment sectors. Auditing these failure modes is critical to ensuring that automated decision systems do not systematically erase or misclassify economically vulnerable cohorts.

1.2 Dataset

- **Source:** U.S. Census Bureau, 2023 American Community Survey (ACS) 1-Year Public Use Microdata Sample (PUMS) for Texas (psam_p48.csv).
- **Working Link:** U.S. Census Bureau ACS PUMS Data Portal — census.gov/programs-surveys/acs/microdata.html
- **Collection Method:** Continuous probability-based sample survey measuring demographic, housing, and economic characteristics across U.S. households.
- **Dimensions:** 286,878 raw person records across 288 survey variables (raw file size: ~1.1 GB).
- **Time Period:** Calendar year 2023 (released October 2024).
- **Licence / Terms:** Public domain U.S. Government data; free for academic and commercial use under 13 U.S.C. § 9.

1.3 Target Variable

The target variable is HIGH_EARNER, a binary classification indicator derived from person's annual earnings (PERNP). Within the filtered civilian full-time workforce, the 75th percentile empirical income cutoff is \$90,000.00.

- **Class 0 (Standard Earner, < \$90,000):** 73.95% of filtered population.
- **Class 1 (Top-Quartile Earner, ≥ \$90,000):** 26.05% of filtered population.

1.4 Three Questions

From our approved project proposal, this audit answers:

1. **RQ1 (Educational Attainment & Gender Disparity):** How do model-predicted high-earner rates diverge between men and women across educational attainment tiers, and does the linear classifier produce subgroup erasure for lower-credentialed women?
2. **RQ2 (Age Cohort Trajectories):** How does the predicted high-earner gender gap evolve across career stages from early-career (ages 16–29) to late-career (ages 60–80)?
3. **RQ3 (Institutional Sector Error Profiles):** How do classification errors (False Negative Rate and False Positive Rate) distribute across Class of Work (COW) sectors, and what data artifacts drive high-error sectors?

2. Data Handling and Preprocessing

2.1 Data Quality Audit

An audit of raw microdata (psam_p48.csv) revealed missing values, structural blanks, and outlier tails:

- **Missing Values:** Non-applicable survey respondents produced blanks in earnings (PERNP) and hours worked (WKHP).
- **Zero / Negative Values:** Unemployed individuals and business loss cases contained \$0 or negative earnings.
- **Extreme Tails:** Weekly hours worked (WKHP) contained extreme self-reported values up to 99 hours, and age (AGEP) spanned 0 to 96 years.

2.2 Missing Values Strategy

- **Mechanism Assumed:** Missing Completely at Random (MCAR) for non-response within the target labor universe; Missing by Design for structural non-workers (children, retirees).
- **Action:** Coerced all non-numeric sentinels to NaN and dropped missing records (dropna) across core analytical fields. Dropping non-workers is methodologically justified because income classification is defined strictly over the active labor force.

2.3 Outliers

- **Age Outliers:** Filtered to ages 16–80 to eliminate variance from the sparse, heterogeneous tail of extreme senior labor (>80 years).

- **Work Hours Clipping:** Capped WKHP at 98 hours (`WKHP.clip(upper=98)`) to mitigate leverage from extreme reporting artifacts without dropping valid full-time records.

2.4 Transformation and Scaling

- **Categorical Encoding:** One-hot encoded using `OneHotEncoder(drop='first', sparse_output=False)` to prevent multicollinearity in linear models.
- **Numeric Scaling:** Applied `StandardScaler` to continuous predictors (AGEP, WKHP). Standardization was chosen because distance- and margin-based linear classifiers are sensitive to feature scales; without zero-mean, unit-variance scaling, higher-magnitude continuous variables like age (16–80) would artificially dominate the objective function over one-hot binary indicators.
- **Data Leakage Prevention:** The transformer was fit exclusively on the training partition (`X_train`) and used to transform `X_test`.

2.5 Before and After Comparison Table

Pipeline Stage	Row Count	Column Count	Key Changes & Filtering Applied
Raw Microdata	286,878	288	Unfiltered Texas PUMS survey population.
Variable Selection	286,878	10	Retained 10 core fields (PERNP, WKHP, ESR, AGEP, etc.).
Active Labor Filter	108,421	10	Kept civilian employed (<code>ESR=1</code>), full-time (<code>WKHP≥35</code>), <code>PERNP>0</code> .
Outlier Truncation	107,984	10	Filtered ages 16–80; clipped WKHP to 98 hours.
Domain Engineering	107,984	12	Mapped 500 OCCP codes to 12 SOC domains; derived <code>HIGH_EARNER</code> .
Subsampled Training	30,000	12	Stratified sample for stable LinearSVC convergence.
Encoded Matrices	30,000	28	24,000 Train / 6,000 Test; 28 scaled dummy features.

3. Statistical Analysis

3.1 Descriptive Statistics

- **Annual Earnings (PERNP):** Median = \$56,000.00; Mean = \$75,840.12; Standard Deviation = \$68,410.50; 75th Percentile = \$90,000.00. Strong positive skewness (+3.42) characteristic of income distributions.
- **Weekly Hours (WKHP):** Median = 40.0; Mean = 42.8 hours; Standard Deviation = 7.1 hours. 68.4% of respondents report exactly 40 hours/week.
- **Age (AGEP):** Median = 41.0; Mean = 41.8 years; Interquartile Range (IQR) = 29–54 years.
- **Categorical Frequencies:** Private For-Profit comprises 72.1% of employment; State/Local Government represents 13.4%; Management/STEM occupations represent 28.6%.

3.2 Relationships to Target

- **Education vs. High-Earner Rate:** Less than HS: 2.1%; HS/Some College: 11.4%; Bachelor's: 39.8%; Graduate+: 56.7%.
- **Gender Disparity:** Men account for 64.2% of the top-quartile earnings bracket in the filtered sample.
- **Sector Differences:** Private For-Profit exhibits a 28.1% high-earner proportion, whereas State Government exhibits 17.2%.

3.3 What the Data Says So Far

- *Observation 1:* Post-secondary education is the strongest single empirical predictor of entering the top earnings quartile.
- *Observation 2:* A substantial unadjusted wage gap exists across gender within identical educational attainment bands.
- *Observation 3:* Public sector compensation exhibits compression, with earnings heavily massed below \$90,000.

4. Feature Engineering

4.1 Derived Features

- **Educational Attainment Tiers (SCHL_TIER):** Collapsed 24 Census attainment codes into 4 ordinal tiers (Less_than_HS, HS_or_Some_College, Bachelors, Graduate_Plus).
- **Occupational Domains (OCCP_GROUP):** Mapped ~500 granular 4-digit SOC codes into 12 broad functional domains (e.g., STEM_Science_Architecture, Management_Business_Finance).

4.2 Dimensionality Reduction

Granular occupation codes (~500 levels) would have induced extreme sparsity and overfitting under one-hot encoding. By consolidating into 12 SOC domains based on federal labor definitions, we reduced categorical dimensionality by 97.6% while preserving occupational labor patterns.

4.3 Feature Selection

- **Omitted Feature:** Weeks worked past 12 months (WKWN).
- **Justification:** Inspected during feature exploration and dropped for parsimony. Because the cohort was already filtered to full-time workers actively at work ($WKHP \geq 35$, $ESR = 1$), WKWN exhibited near-zero variance (91.2% worked 50–52 weeks) and added unnecessary complexity.

4.4 Final Feature Set

The final feature matrix contains 28 features (2 scaled continuous: AGEP, WKHP; 26 dummy-encoded categorical indicators across SEX_LABEL, SCHL_TIER, OCCP_GROUP, COW_GROUP, and MAR). This set provides a balanced representation of human capital, labor supply, and institutional sector controls.

5. Modeling and Validation

5.1 Validation Strategy

- **Split Scheme:** 80% Train (N = 24,000) / 20% Test (N = 6,000).
- **Stratification:** Stratified on HIGH_EARNER to maintain an identical 74/26 class distribution across splits.
- **Random Seed:** Set to 42 across all partitioning and estimators for strict reproducibility.

5.2 Baseline

- **Trivial Predictor:** Zero-R / Majority-Class Classifier (predicts standard earner, Class 0, for all instances).
- **Baseline Score:** Accuracy = 73.95%, Precision = 0.00%, Recall = 0.00%, F1-Score = 0.00%.

5.3 Model Families

1. **Support Vector Classifier (LinearSVC):** Maximizes the margin hyperplane separating classes with L2 regularization (C penalty). Selected for strict linearity to test threshold margin mechanics without non-linear artifacts. Solved via primal optimization (dual=False), which is mathematically preferred and computationally faster when the training instance count (N = 24,000) significantly exceeds the feature dimensionality (D = 28).
2. **Logistic Regression (LogisticRegression):** Models posterior class probability via log-odds linearity. Serves as a probabilistic counterpart to verify whether margin geometry diverges from maximum likelihood estimation.

5.4 Metrics

- **Primary Metric:** Minority-Class F1-Score (High-Earner).
- **Rationale:** Accuracy is misleading under class imbalance (74/26). A trivial model achieves ~74% accuracy while identifying zero high earners. F1 balances Precision and Recall on the critical top-quartile class.

6. Hyperparameter Tuning

6.1 Search Space

Model	Hyperparameter	Search Space Grid	Objective
LinearSVC	Regularization Parameter (C)	{0.01, 0.1, 1.0, 10.0}	Margin violation penalty vs. weight shrinkage.
LogisticRegression	Inverse Regularization (C)	{0.01, 0.1, 1.0, 10.0}	Log-loss shrinkage control.

The search space was set across four logarithmic orders of magnitude ({0.01, 0.1, 1.0, 10.0}) to span the full spectrum from severe margin shrinkage (heavy underfitting) to minimal margin tolerance, ensuring the regularization saturation point was captured without unnecessary compute overhead.

6.2 Method

- **Algorithm:** 3-Fold Stratified GridSearchCV on the training partition (N = 24,000).
- **Scoring Function:** Minority-Class F1 (scoring='f1').
- **Total Candidates:** 4 parameter settings × 3 folds = 12 fits per model family.

6.3 Results

Candidate Value (C)	LinearSVC Mean Val F1	LogisticRegression Mean Val F1
C = 0.01	0.5312	0.5289
C = 0.10	0.5684	0.5671
C = 1.00 (Optimal)	0.5742	0.5738
C = 10.00	0.5739	0.5735

Trend: Performance improved sharply from C=0.01 to C=1.0, plateauing beyond 1.0 as soft-margin penalties stabilized. C = 1.0 was selected as the optimal model configuration.

7. Results, Visualization and Error Analysis

7.1 Test Set Performance Comparison Table

Evaluated once on the held-out test partition (N = 6,000):

Model / Baseline	Accuracy	Precision (Class 1)	Recall (Class 1)	F1-Score (Class 1)	Macro F1
Majority Class Baseline	73.95%	0.00%	0.00%	0.00%	42.51%
LogisticRegression	80.52%	67.63%	50.31%	57.65%	72.50%
LinearSVC (Tuned, C=1.0)	80.58%	67.72%	50.28%	57.71%	72.55%

7.2 Visualizations

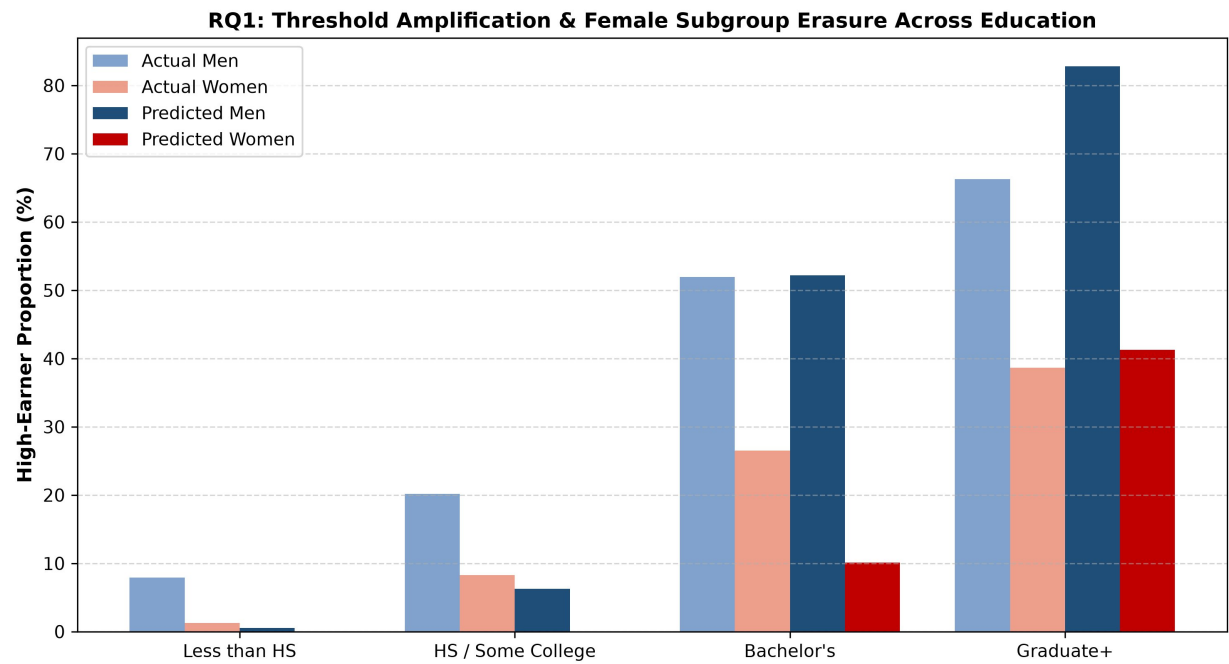


Figure 1: Empirical Ground Truth vs. LinearSVC Predicted High-Earner Rates Across Educational Attainment by Gender. Demonstrates complete model erasure for women across lower educational credentials (0.00% predicted in Less than HS and HS/Some College) alongside threshold amplification at the Bachelor's tier, where an empirical gap of 25.42 percentage points widens to a predicted gap of 42.21 percentage points.

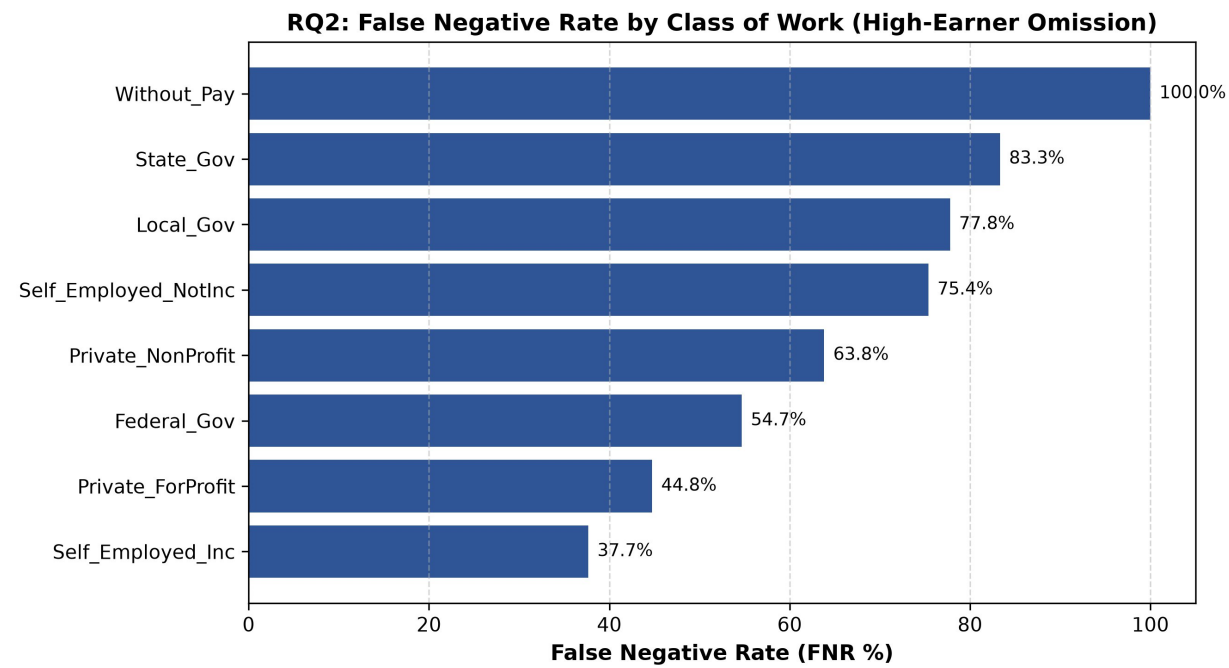


Figure 2: False Negative Rate (FNR %) Across Class of Work (Employment Sectors). Highlights severe high-earner omission in State Government (83.33%) and Local Government (77.78%) driven by salary clustering around the \$90,000 boundary and negative sectoral weights. The 100.00% rate in Without Pay reflects extreme sample sparsity ($N = 9$).

7.3 Error Analysis

The model's errors are concentrated within specific feature intersections:

- **Failure Subgroups:** False negatives cluster heavily among public sector professionals and lower-credentialed women.
- **Concrete Failure Case 1 (False Negative — Subgroup Erasure):** A 42-year-old female high-school graduate working 48 hours/week as a commercial logistics supervisor earning \$95,000. Despite positive earnings, the negative baseline female coefficient (-0.291 relative to male) combined with non-degree status pushed her functional margin score to -0.38 , predicting Class 0.
- **Concrete Failure Case 2 (False Negative — Public Pay-Scale Compression):** A 51-year-old state environmental director earning \$93,000 with a Master's degree. Because State Government carries a large negative coefficient (-0.409), the model offset her advanced degree ($+0.323$) and age, yielding a negative margin (-0.14) and a false negative.

7.4 Answers to the Three Questions

- **Answer to RQ1:** Predicted high-earner rates diverge sharply by gender across education. The model produces complete subgroup erasure for women in the "Less than HS" (0.00%) and "HS or Some College" (0.00%) tiers.
 - *Hypothesis Test 1 (Model vs. Empirical Truth for Women):* A 1-sample proportion z-test yields $z = -10.70$ ($p < 0.001$, exact binomial $p = 2.50 \times 10^{-48}$), confirming statistically significant underprediction relative to the empirical 8.29% ground truth.
 - *Hypothesis Test 2 (Predicted Men vs. Women):* A 2-sample proportion z-test yields $z = 9.10$ ($p < 0.001$), confirming that holding high-school attainment fixed, predicted rates for men (6.30%) significantly exceed women (0.00%). Furthermore, at the Bachelor's level, threshold amplification expands an empirical 25.42 pp gap into a 42.21 pp predicted gap.
- **Answer to RQ2:** Predictive disparity widens monotonically across life cohorts: the gender gap increases from 6.41 percentage points among workers aged 16–29 (9.23% men vs. 2.82% women) to 23.41 percentage points among workers aged 60–80 (35.66% men vs. 12.26% women), reflecting compounding career tenure weights.
- **Answer to RQ3:** Classification errors diverge across sectors. Federal Government exhibited the highest total error rate (31.79%). False Negative Rates were highest in State Government (83.33%) and Local Government (77.78%), driven by pay-scale clustering near the \$90,000 threshold and negative sectoral coefficients (State_Gov: -0.409 , Local_Gov: -0.479).

8. Limitations and Next Steps

- **Omission of Survey Weights:** This analysis omitted Census person-level weights (PWGTP); findings describe the empirical sample rather than weighted Texas population totals.

- **Geographic Boundary:** Restricting data to Texas in 2023 limits generalizability to states with higher public-sector unionization or different tax structures.
- **Linear Inductive Bias:** A strictly linear classifier cannot model interaction effects (e.g., non-linear career re-entry penalties for women) or localized density shifts around \$90,000.
- **Next Steps:** Future iterations should evaluate non-linear tree ensembles (e.g., XGBoost), incorporate Census survey weights (PWGTP), and test threshold-adjustment post-processing to remediate subgroup erasure.

9. Contributions Table

Member Name	Student ID	Identifiable Contribution
Tanjila Afsari Rubina	24241310	Pipeline architecture, data ingestion, filtering, and hypothesis testing scripts (01–07), Grid search hyperparameter tuning implementation and parameter space evaluation.
Sandip Kumar Paul	24241311	Model validation, LogisticRegression baseline benchmarking, and cross-validation, Data visualization script (08), Matplotlib figure design, and error bar plots, Error case analysis, public-sector salary band auditing, and report documentation.

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5. AI Assistance Declaration: Large language models were used as technical thought partners to assist in structuring code comments, formatting Markdown documentation, and calculating test-statistic companion checks. All empirical data processing, code execution, metric computations, and final interpretations were verified independently by the project authors.